

Safa Paravakkal

Biomedical Sciences Student · Neuroscience, Digital Health & Research Software

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PROFILE

Third-year Biomedical Sciences student at Royal Holloway with a strong interest in neuroscience, digital health, and research software. I'm building full-stack tools for searching, analysing, and appraising biomedical research, and have hands-on wet lab experience from a molecular cloning project in *E. coli*. I'm seeking internships or placements where I can contribute to health tech, research, or data-driven product teams.

PROJECTS

NeuroTrials Explorer – Data Dashboard · Solo Project *April 2026 · Completed*

An interactive Streamlit dashboard exploring clinical trial trends across 7 neurological conditions using the ClinicalTrials.gov API · Python · Requests · Pandas · Streamlit · Plotly

- Designed and iterated on an ETL workflow for clinical trial records across 7 neurological conditions
- Used Python and pandas to clean trial data
- Implemented and customised 6 interactive Plotly Express charts to explore trial patterns by condition, phase, sponsor, status, and year
- Deployed on Streamlit Community Cloud

Medical Evidence Platform (Seren) – Solo Project *March 2026 · Live and under active development*

A full-stack platform for discovering, importing, and AI-analysing peer-reviewed medical research · Python · FastAPI · SQL Model · SQLite · pytest · GitHub Actions

- Built a unified search workflow across Europe PMC, OpenAlex, Crossref, and Semantic Scholar
- Developed provider parsing and deduplication logic using DOI matching, metadata comparison, and cross-source reference tracking
- Implemented search filters for study type, including RCTs, meta-analyses, cohort studies, and systematic reviews
- Added AI-powered research analysis features with model routing and fallback logic
- Maintained a 93-test suite covering provider parsing, authentication, deduplication, and study workflows
- Set up CI/CD with GitHub Actions to run tests automatically on each push

Term 3 Molecular Lab Project – Royal Holloway, University of London *May – June 2026 · Completed*

- Cloned a 1386 bp fragment of DISP1 (Hedgehog pathway regulator) into a pLATE52 expression vector using Ligation-Independent Cloning; full pipeline from primer design and gradient PCR through bacterial transformation (DH5α/BL21) and miniprep/midiprep
 - Verified recombinant protein expression via IPTG induction, SDS-PAGE, and Western blotting
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EXPERIENCE

Sales Advisor – H&M — Sloane Square & Kensington *August 2024 – June 2025*

- Customer service in a high-footfall flagship store, maintaining standards during regional management visits
- Supported stock control, replenishment, and visual merchandising
- Built strong communication, reliability, and time-management skills through continuous customer-facing work

Virtual Work Experience – DigData *May 2026*

- Analysed a simulated clinical trial dataset; cleaned and merged patient data, engineered a BMI variable, and identified protein concentration as a potential predictive biomarker for treatment response.

Volunteer – Nettlestead Care Home *May 2023*

- Supported residents and staff, gaining exposure to patient-facing communication and care settings

Wellbeing Secretary – Biology Society of Royal Holloway *June 2026 - current*

EDUCATION

BSc Biomedical Sciences — Royal Holloway, University of London 2024 – Present · Third Year

A Levels — Bullers Wood School for Girls 2022 – 2024

Chemistry · Biology · Psychology

GCSEs · 9 Subjects — Coopers Secondary School 2017 – 2022

Incl. English, Maths, Triple Science, History, Art

SKILLS

Technical: Python · pandas · Streamlit · Plotly · REST APIs · FastAPI · SQLAlchemy · SQLite · pytest · GitHub Actions · HTML/CSS

Scientific & Domain: Clinical evidence hierarchy · RCT & meta-analysis appraisal · PICO framework · Biomedical research methods · Medical literature databases · Clinical trial data analysis

Laboratory & Wet Lab: Ligation-Independent Cloning (LIC) · Gradient PCR · Bacterial transformation (DH5α/BL21) · SDS-PAGE · Western blotting · IPTG induction · Nanodrop quantification · Aseptic technique

Professional: Independent project delivery · Analytical problem-solving · Team collaboration · Time management

ADDITIONAL

Certifications: Google Data Analytics — Coursera (in progress)

Availability: Immediately available — internships, work experience, short-term placements